THE ROWS OF THE DIVISIBLE-PRODUCT TRIANGLE OEIS A090905 END AT TWICE THE BERTRAND PRIMES

MATH0

Abstract. Split $1, 2, 3, \ldots$ into consecutive blocks: block 1 is (1), and each later block is the shortest block, not shorter than its predecessor, whose product is a multiple of the product of its predecessor (OEIS A090905). We prove Amarnath Murthy's 2003 conjecture from this entry: for every $n > 4$ the last term of block n is $2p$, where p is the largest prime of block $n-1$, and p is the Bertrand prime B_{n-1} (OEIS A006992: $B_1 = 2$, and B_{k+1} is the largest prime below $2B_k$). The proof combines Legendre's formula with a prime in $(5x/3, 2x)$ for every integer $x \geq 83$, a weak form of Nagura's 1952 theorem, which we reprove by an explicit prime ladder and Chebyshev's method. We know of no earlier proof, so we classify the result as new, although the argument is elementary once the prime-gap bound is available. The theorem is verified in Lean 4 against the frozen, fidelity-reviewed math0 formalization of the conjecture, whose definitions are checked by the kernel against the OEIS data.

1. Introduction

OEIS A090905 [8], contributed by Amarnath Murthy on December 13, 2003, is the "Left side of irregular triangle of natural numbers in which every row product is a multiple of the previous". The triangle splits $1, 2, 3, \ldots$ into consecutive blocks ("groups") (1), (2), $(3, 4)$, $(5, \ldots, 8)$, $(9, \ldots, 14)$, $(15, \ldots, 26)$, $(27, \ldots, 46)$, ... Each block starts right after the previous one, and its length is the least k, not smaller than the previous length, such that the product of the previous block divides the product of the new one; this is the rule of the Mathematica program of M. De Vlieger (2016) in the entry. The companion entries A090904, A090906 and A090907 [7] list the row products, the row lengths, and the ratios of consecutive row products. All four entries carry Murthy's conjecture (quoted in Section 2): for $n > 4$ the last term of row n is twice the largest prime of row $n-1$, and these primes are the Bertrand primes A006992 [6]. We prove it (Theorem 2).

Method. Let $b < a < c$ be consecutive Bertrand primes and suppose row n is $[2b+1, 2a]$. The prime c lies in row n, and no multiple of c lies in $(2a, 2c)$, so row $n+1$ must reach $2c$; ending at $2c$ suffices because $(2a)!^2 \mid (2b)!\,(2c)!$ (Legendre's formula). Both steps use $3a > 5b$ and $3c > 5a$, i.e. a prime in $(5x/3, 2x)$ for every integer $x \geq 83$. This follows from Nagura's theorem [5] (a prime in $(x, 6x/5)$ for $x \geq 25$); the Lean development proves the needed form from scratch, by a ladder of primes below 10^6 and by Chebyshev's method [1] above.

Status and prior work. The entries A090904–A090907 (last edited in December 2016) state the claim only as a conjecture. A090907 [7] gives a formula for the ratios in terms of A006992, "verified for $4 < n < 21$". A090906 [7] contains an unproved formula of M. Bouayoun (March 2004) for the row lengths, $2(\text{A006992}(n) - \text{A006992}(n-1))$ for $n > 4$; as written it is off by one index (it gives 12 for $n = 5$, whereas the fifth row length is 6). Theorem 2 gives the shifted form $2(\text{A006992}(n-1) - \text{A006992}(n-2))$ for $n \geq 6$, and the formula of A090907 for $n \geq 6$ (neither consequence is formalized). A381901 [10] (A. Niedermaier, March 2025) defines the same partition without the lower bound on the row length and states, without proof,

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that the last term of row n is $2\,\mathrm{A006992}(n-1)$ for $n \geq 5$; this follows from Theorem 2 and Remark 3. Related entries are A113117, A113118 (L. Quet, 2006) and Murthy's variant A093911 [9]. The math0 referees found no proof or counterexample in searches of arXiv, MathOverflow, Mathematics Stack Exchange and SeqFan-style queries, and no formalization in the formal-conjectures repository [2].

2. Statement

For integers $s \geq 1$ and $k \geq 0$ write $s^{\overline{k}} = s(s+1)\cdots(s+k-1)$ (Lean's `Nat.ascFactorial`), the product of the block $s, \ldots, s+k-1$.

Definition 1. Let $a(1) = 1$, $L(1) = 1$ and, for $n \geq 1$,

$$a(n+1) = a(n) + L(n), \qquad L(n+1) = \min\{k \geq L(n) : a(n)^{\overline{L(n)}} \mid a(n+1)^{\overline{k}}\}.$$

Row n is the block $[a(n), e(n)]$ with $e(n) = a(n) + L(n) - 1$. The Bertrand primes are $B_1 = 2$ and $B_{k+1} =$ the largest prime $< 2B_k$; thus $(B_k)_{k\geq 1} = 2, 3, 5, 7, 13, 23, 43, 83, 163, 317, \ldots$

The minimum exists: $k = a(n+1) - 1 = a(n) + L(n) - 1 \geq L(n)$ is admissible, because the row product divides $(a(n+1)-1)!$, which divides the product of any $a(n+1)-1$ consecutive integers. The COMMENTS section of A090905 reads, verbatim (the entry is by Amarnath Murthy, December 13, 2003):

> `Conjecture: For n > 4 the last term of the n-th group is 2p where p is the largest prime in the (n-1)th group. And these are the Bertrand primes.`

Theorem 2. *For every $n \geq 5$, the set of primes in row $n-1$ has a largest element p; moreover $e(n) = 2p$ and $p = B_{n-1}$.*

Applying this to n and to $n-1$ and using $a(n) = e(n-1) + 1$, we see that row n is $[2B_{n-2}+1,\, 2B_{n-1}]$ and $L(n) = 2B_{n-1} - 2B_{n-2}$ for every $n \geq 6$.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A090905.lean`, reviewed for fidelity and frozen (SHA-256 `8616fe97...a42a44`) before any proof attempt. Its definitions read as follows (verbatim; docstrings and the auxiliary theorems are omitted):

```
namespace Math0.Statements.A090905
def search (P s : ℕ) : ℕ → ℕ → ℕ → ℕ
  | 0, k, _ => k
  | fuel + 1, k, Q => if P ∣ Q then k else search P s fuel (k + 1) ((s + k) * Q)
def next (r : ℕ × ℕ) : ℕ × ℕ :=
  (r.1 + r.2,
    search (r.1.ascFactorial r.2) (r.1 + r.2) (r.1 + r.2) r.2 ((r.1 + r.2).ascFactorial r.2))
def row (n : ℕ) : ℕ × ℕ := next^[n - 1] (1, 1)
def a (n : ℕ) : ℕ := (row n).1
def len (n : ℕ) : ℕ := (row n).2
def last (n : ℕ) : ℕ := a n + len n - 1
def bertrand (n : ℕ) : ℕ := (fun b => Nat.findGreatest Nat.Prime (2 * b - 1))^[n - 1] 2
def statement : Prop :=
  ∀ n : ℕ, 4 < n → ∃ p : ℕ,
    IsGreatest {q : ℕ | q.Prime ∧ a (n - 1) ≤ q ∧ q ≤ last (n - 1)} p ∧
      last n = 2 * p ∧ p = bertrand (n - 1)
```

A row is stored as the pair (first term, length). Called with $Q = s^{\overline{k}}$, `search P s fuel k Q` returns the least $j \in [k, k+\mathit{fuel})$ with $P \mid s^{\overline{j}}$ (or $k+\mathit{fuel}$ if there is none). The file proves `a_one`, `len_one`,

`a_succ` and `len_succ_isLeast`, i.e. the recursion of Definition 1 for $n \geq 1$; in particular the fuel $a(n+1)$ always suffices (the admissible $k = a(n+1) - 1$ lies in the search window), so `a`, `len`, `last` are exactly a, L, e (and $a(n) \geq 1$, so the truncated subtraction in `last` is harmless). Since $B_k \geq 2$, `Nat.findGreatest Nat.Prime (2 * b - 1)` at $b = B_k$ is the largest prime $< 2B_k$, so `bertrand` is A006992. Lemmas proved by `decide +kernel` check $a(1), \ldots, a(18)$, $L(1), \ldots, L(17)$ and $B_1, \ldots, B_{13}$ against the data of A090905, A090906 and A006992 (`a_data`, `len_data`, `bertrand_data`).

"The largest prime in the $(n-1)$th group" is the greatest element of the set of primes in $[a(n-1), e(n-1)]$, whose existence is asserted. "These are the Bertrand primes" is read as $p = B_{n-1}$: for $n = 5$ the largest prime of row $4 = (5, \ldots, 8)$ is $7 = B_4$, and the rows are disjoint and increasing ($2, 3, 5$ do not occur as such p). This is the strongest natural reading; the weaker reading "each such p is a Bertrand prime" follows from it.

Remark 3. The bound $k \geq L(n)$ comes from De Vlieger's program, not from the entry's name. Dropping it does not change the triangle (not formalized): rows 1–20 agree by computer check, and for $n \geq 10$ the minimality argument of Lemma 12 does not use the bound.

3. Proof

We follow the Lean proof; names of its (private) lemmas are given in brackets.

3.1. A prime between $5x/3$ and $2x$.

Proposition 4 (`exists_prime`)**.** *For every integer $x \geq 83$ there is a prime p with $5x < 3p < 6x$.*

This follows from Nagura's theorem [5] at $5x/3 \geq 25$; we give the self-contained proof that is formalized.

Lemma 5 (`ladder_aux`)**.** *Let $q_0 = 138$, and let $q_1 < \cdots < q_{53}$ be the primes 163, 193, 229, …, 1491013, 1789201 of the list `ladder`, which satisfy $5q_i < 6q_{i-1}$ for $1 \leq i \leq 53$. Then Proposition 4 holds for $83 \leq x < 10^6$.*

Proof. We have $3q_0 = 414 < 5x$ and $5x < 5 \cdot 10^6 < 3q_{53} = 5367603$. Let $i \geq 1$ be least with $5x < 3q_i$. Then $3q_{i-1} \leq 5x$, so $15q_i < 18q_{i-1} \leq 30x$, i.e. $3q_i < 6x$. In Lean, `norm_num` produces a primality certificate for each q_i, which the kernel checks, and the 53 inequalities $5q_i < 6q_{i-1}$ are checked by `decide`. □

Let Λ be the von Mangoldt function, $\psi(y) = \sum_{d \leq y} \Lambda(d)$ and $\theta(y) = \sum_{p \leq y} \log p$.

The identity $\log M! = \sum_{d \leq M} \Lambda(d) \lfloor M/d \rfloor$ follows by induction on M from $\lfloor (M+1)/d \rfloor - \lfloor M/d \rfloor = [d \mid M+1]$ and $\sum_{d \mid n} \Lambda(d) = \log n$; with $\lfloor \lfloor N/k \rfloor / d \rfloor = \lfloor \lfloor N/d \rfloor / k \rfloor$ it gives $\log \lfloor N/k \rfloor ! = \sum_{d \leq N} \Lambda(d) \lfloor \lfloor N/d \rfloor / k \rfloor$ for $k \geq 1$ (`log_factorial_div_eq_sum`). A *weight* is a list W of pairs $(c_i, k_i) \in \mathbb{Z} \times \mathbb{N}_{\geq 1}$. Put $H_W(n) = \sum_i c_i \lfloor n/k_i \rfloor$ and $G_W(N) = \sum_i c_i \log \lfloor N/k_i \rfloor !$. Then

$$G_W(N) = \sum_{d \leq N} \Lambda(d)\, H_W(\lfloor N/d \rfloor). \tag{1}$$

Lemma 6 (`Gw_le_psi`, `psi_le_Gw`)**.** (a) *If $H_W \leq 1$ on $\mathbb{N}$, then $G_W(N) \leq \psi(N)$.* (b) *If $H_W \geq 0$ on $\mathbb{N}$ and $H_W(n) \geq 1$ whenever $P \nmid n$, then $\psi(N) \leq G_W(N) + \psi(\lfloor N/P \rfloor)$.*

Proof. (a) is immediate from (1) and $\Lambda \geq 0$. For (b), split $\psi(N) = \sum_{d \leq N} \Lambda(d)$ according to whether $P \mid \lfloor N/d \rfloor$. If so, then $P \leq \lfloor N/d \rfloor$ (as $\lfloor N/d \rfloor \geq 1$), hence $d \leq \lfloor N/P \rfloor$, and these terms contribute at most $\psi(\lfloor N/P \rfloor)$. The other terms satisfy $\Lambda(d) \leq \Lambda(d) H_W(\lfloor N/d \rfloor)$, and the remaining terms of (1) are ≥ 0. □

We use Chebyshev's weight [1] and a 19-term weight found by linear programming over the divisors of 420:

$$W_L = (1,1),(-1,2),(-1,3),(-1,5),(1,30),$$
$$W_U = (1,1),(-1,2),(-1,3),(-1,5),(1,6),(-1,7),(1,10),(-1,12),(1,14),(-1,20),$$
$$(-1,30),(1,35),(-1,42),(1,60),(-1,70),(1,84),(-1,105),(1,210),(-4,420).$$

If every k_i divides P, then $H_W(n) = H_W(n \bmod P) + \lfloor n/P \rfloor\, R_W(P)$ with $R_W(P) = \sum_i c_i P/k_i$ (`Hw_mod`). Here $R_{W_L}(30) = R_{W_U}(420) = 0$, so H_{W_L} has period 30 and H_{W_U} has period 420. Evaluating one period in the kernel (`decide` in `HL_le`, `decide +kernel` in `HU_spec`) gives $H_{W_L} \le 1$, $H_{W_U} \ge 0$, and $H_{W_U}(n) \ge 1$ for $420 \nmid n$. Lemma 6 then gives $G_{W_L}(N) \le \psi(N) \le G_{W_U}(N) + \psi(\lfloor N/420 \rfloor)$.

Lemma 7 (`log_fact_lower`, `log_fact_upper`)**.** *For $M \ge 1$: $M\log M - M + \frac{1}{2}\log M + \frac{1}{2} \le \log M! \le M\log M - M + \frac{1}{2}\log M + 1$.*

Proof. The lower bound is `Stirling.le_log_factorial_stirling` with $\log(2\pi) \ge 1$. For the upper bound, $M!/(\sqrt{2M}\,(M/e)^M)$ is non-increasing in $M \ge 1$ (`Stirling.log_stirlingSeq'_antitone`) and equals $e/\sqrt{2}$ at $M = 1$. □

Lemma 8 (`Gw_bounds`, `psi_lower`, `psi_upper`)**.** *For every integer $m \ge 1$, $386.94\,m - 5.025 - \frac{1}{2}\log m \le \psi(420m) \le 448.99\,m - 3.685 - 3\log m + \psi(m)$.*

Proof. Let W be W_L or W_U and $r_i = 420/k_i \in \mathbb{N}$, so that $\lfloor 420m/k_i \rfloor = r_i m$. Apply Lemma 7 to $M = r_i m$, write $\log M = \log r_i + \log m$, multiply by c_i (reversing the inequalities when $c_i < 0$) and sum. Since $\sum_i c_i r_i = 0$, the terms in $m\log m$ and m cancel, and

$$mD_1 + \tfrac{1}{2}D_2 + \tfrac{S}{2}\log m + E^- \;\le\; G_W(420m) \;\le\; mD_1 + \tfrac{1}{2}D_2 + \tfrac{S}{2}\log m + E^+,$$

where $D_1 = \sum c_i r_i \log r_i$, $D_2 = \sum c_i \log r_i$, $S = \sum c_i$, $E^+ = \sum_{c_i>0} c_i + \sum_{c_i<0} c_i/2$ and $E^- = \sum_{c_i>0} c_i/2 + \sum_{c_i<0} c_i$. For W_L: $D_1 = 196\log 2 + 126\log 3 + 70\log 5 \ge 386.94$, $D_2 = -\log 420 \ge -6.05$, $S = -1$, $E^- = -2$. For W_U: $D_1 = 192\log 2 + 123\log 3 + 70\log 5 + 35\log 7 \le 448.99$, $D_2 = -\log 11760 \le -9.37$, $S = -6$, $E^+ = 1$. These numerical bounds use Mathlib's $0.6931471803 < \log 2 < 0.6931471808$ and 16 terms of the series of $\log(1-t)$ at $t = \frac{1}{4}, \frac{1}{5}, \frac{1}{8}$ with Mathlib's remainder bound `Real.abs_log_sub_add_sum_range_le`. Combining with $G_{W_L}(420m) \le \psi(420m) \le G_{W_U}(420m) + \psi(m)$ gives the claim. □

Proof of Proposition 4 for $x \ge 10^6$ (`exists_prime_large`). For an integer p, $5x < 3p < 6x$ means $\lfloor 5x/3 \rfloor < p \le 2x - 1$. Suppose there is no such prime; then $\theta(2x-1) = \theta(\lfloor 5x/3 \rfloor)$. Let $m_0 = \lfloor (2x-1)/420 \rfloor$, $m_1 = \lfloor \lfloor 5x/3 \rfloor / 420 \rfloor + 1$, $s = \sqrt{x} \ge 1000$ and $\lambda = \log x$. Then $1 \le m_0, m_1 \le x$, $420m_0 \le 2x-1$, $2x \le 420m_0 + 420$, $\lfloor 5x/3 \rfloor \le 420m_1$, $1260m_1 \le 5x + 1260$ and $196m_1 \le x$. We use Mathlib's bounds $\psi(y) - \theta(y) \le 2\sqrt{y}\log y$ and $\psi(y) \le y\log 4 + 2\sqrt{y}\log y$ for $y \ge 1$, the monotonicity of ψ, $\theta \le \psi$, Lemma 8, $\sqrt{2x-1} \le 1.4143\,s$, $\log(2x-1) \le \log 2 + \lambda$, $0 \le \log m_i \le \lambda$ and $\sqrt{m_1} \le s/14$:

$$\theta(2x-1) \ge \psi(420m_0) - 2\sqrt{2x-1}\log(2x-1) \ge \frac{386.94\,x}{210} - 391.965 - \frac{\lambda}{2} - 2.8286\,s(\log 2 + \lambda),$$
$$\theta(\lfloor 5x/3 \rfloor) \le \psi(420m_1) \le \frac{448.99\,x}{252} + 445.305 + \frac{1.3864\,x}{196} + \frac{s\lambda}{7}.$$

The left sides are equal, $386.94/210 - 448.99/252 - 1.3864/196 > 0.05379$ and $2.8286\log 2 \le 1.9608$, so $0.05379\,x \le 837.27 + \lambda/2 + 1.9608\,s + 2.9715\,s\lambda$. From $\log(s/1000) \le s/1000 - 1$ and $\log 1000 = 3\log 2 + 3\log 5 \le 6.9078$ we get $\lambda = 2\log s \le 11.8157 + s/500$. Substituting this and $x = s^2$ yields $0.0478\,s^2 - 37.08\,s - 843.2 \le 0$, which is false because $s^2 \ge 1000s$ (one `linarith` call in Lean). □

3.2. **The Bertrand chain.**

Lemma 9 (`chain_step`, `bertrand_ge`)**.** *For $k \geq 8$: $B_k \geq 83$, $5B_k < 3B_{k+1}$, $B_{k+1} < 2B_k$, and no prime lies in $(B_{k+1}, 2B_k)$.*

Proof. If $b \geq 83$ and c is the largest prime $\leq 2b-1$, Proposition 4 gives a prime $p \leq 2b-1$ with $5b < 3p$, so $5b < 3c$; the other claims hold by the choice of c. Now induct from $B_8 = 83$ (a kernel computation). □

3.3. **The factorial divisibility.**

Lemma 10 (`pointwise`)**.** *Let $a < 2b$ and $5a < 3c$. For an integer $y \geq 1$ put $k = \lfloor 2a/y \rfloor$ and $f(y) = \lfloor 2b/y \rfloor + \lfloor 2c/y \rfloor - 2k$. Then* (i) *$f(y) \geq -1$;* (ii) *$f(y) \geq 0$ unless $c < y \leq 2a$;* (iii) *$f(y) \geq 1$ if $k \geq 13$.*

Proof. Let $q_1 = \lfloor 2b/y \rfloor$, $q_2 = \lfloor 2c/y \rfloor$. From $ky \leq 2a < 4b < 2(q_1+1)y$ and $5ky \leq 10a < 6c < 3(q_2+1)y$ we get $2q_1 \geq k-1$ and $3q_2 \geq 5k-2$, so $q_1 + q_2 \geq (13k-7)/6$, which is $\geq 2k-1$ for $k \geq 1$ and $\geq 2k+1$ for $k \geq 13$. This gives (iii), and (i) since $f \geq 0$ when $k = 0$. For (ii): $y > 2a$ forces $k = 0$. If $y \leq c$, then $q_2 \geq 2$, which gives $f \geq 0$ for $k \leq 1$; for $k \geq 2$ the bounds $q_1 \geq \lceil (k-1)/2 \rceil$ and $q_2 \geq \lceil (5k-2)/3 \rceil$ give $q_1 + q_2 \geq 2k$ (directly for $2 \leq k \leq 6$, and from $(13k-7)/6 \geq 2k$ for $k \geq 7$). In Lean all cases are closed by `omega`. □

Lemma 11 (`fact_dvd`)**.** *Let $b \geq 83$, $5b < 3a$, $a < 2b$, $5a < 3c$, $c < 2a$, and suppose that no prime lies in $(c, 2a)$. Then $(2a)!^2 \mid (2b)!\,(2c)!$.*

Proof. By Legendre's formula (Mathlib's `Nat.factorization_factorial`) it suffices to show $\sum_{j \geq 1} f(\ell^j) \geq 0$ for every prime ℓ, with f as in Lemma 10. If no power ℓ^j ($j \geq 1$) lies in $(c, 2a]$, every term is ≥ 0 by Lemma 10(ii). Otherwise let $c < \ell^j \leq 2a$. This j is unique: if also $c < \ell^i \leq 2a$ with $i < j$, then $2a \geq \ell^j \geq 2\ell^i > 2c$, contradicting $c > 5a/3$. Moreover $j \geq 2$: a prime $\ell \in (c, 2a)$ is excluded by hypothesis, and $\ell = 2a$ is impossible since $2a > 2$ is even. Hence $\ell^2 \leq 2a$, and so $13\ell \leq 2a$ (from $\ell^2 \leq 2a$ if $\ell \geq 13$, and from $2a > 10b/3 > 156$ if $\ell < 13$). Thus $\lfloor 2a/\ell \rfloor \geq 13$, so $f(\ell) \geq 1$ by (iii), $f(\ell^j) \geq -1$ by (i), and all other terms are ≥ 0 by (ii). □

3.4. **The rows.**

Lemma 12 (`step_core`)**.** *Under the hypotheses of Lemma 11, if c is prime, then $2c-2a$ is the least $k \geq 2a-2b$ with $(2b+1)^{\overline{2a-2b}} \mid (2a+1)^{\overline{k}}$.*

Proof. First, $c > 5a/3$ and $b > a/2$ give $b+c > 2a$, i.e. $2c-2a \geq 2a-2b$. Next, $(2b)!\,(2b+1)^{\overline{2a-2b}} = (2a)!$ and $(2a)!\,(2a+1)^{\overline{2c-2a}} = (2c)!$; multiplying the desired divisibility by $(2b)!\,(2a)!$ turns it into Lemma 11, so $k = 2c-2a$ is admissible. Finally, $c > 5a/3 > 25b/9 > 2b$ and $c < 2a$, so the prime c divides $(2b+1)^{\overline{2a-2b}}$. If $k < 2c-2a$, the factors $2a+1, \ldots, 2a+k$ of $(2a+1)^{\overline{k}}$ lie strictly between c and $2c$, so $c \nmid (2a+1)^{\overline{k}}$ and k is not admissible. (This part does not use the bound $k \geq 2a-2b$.) □

Proposition 13 (`rows`)**.** *For every $n \geq 10$, $a(n) = 2B_{n-2} + 1$ and $L(n) = 2B_{n-1} - 2B_{n-2}$.*

Proof. Induction on n. For $n = 10$ the kernel evaluates $a(10) = 167$, $L(10) = 160$, $B_8 = 83$ and $B_9 = 163$ (`decide +kernel`). For the step put $b = B_{n-2}$, $a = B_{n-1}$, $c = B_n$. Lemma 9 gives the hypotheses of Lemmas 11 and 12. By the induction hypothesis row n is $[2b+1, 2a]$, so $a(n+1) = 2a+1$, and `len_succ_isLeast` together with Lemma 12 gives $L(n+1) = 2c-2a$. □

3.5. **Proof of Theorem 2.** For $n = 5, \ldots, 10$, row $n-1$ is $[5,8]$, $[9,14]$, $[15,26]$, $[27,46]$, $[47,86]$, $[87,166]$, with largest prime $p = 7, 13, 23, 43, 83, 163$, which is B_{n-1}, and $e(n) = 14, 26, 46, 86, 166, 326 = 2p$. These values are evaluated by the kernel (`decide +kernel`), and the absence of primes in $(p, e(n-1)]$ is checked by `interval_cases` and `norm_num` (`small_case`). Let $n \geq 11$. By Proposition 13, row $n-1$ is $[2B_{n-3}+1, 2B_{n-2}]$ and $e(n) = 2B_{n-1}$. By the chain inequalities, $p = B_{n-1}$ satisfies $p > 25B_{n-3}/9 > 2B_{n-3}$ and $p < 2B_{n-2}$, so it is a prime in row $n-1$. A prime $q > p$ in row $n-1$ would satisfy $q < 2B_{n-2}$ (as $2B_{n-2} > 2$ is even), contradicting the absence of primes in $(B_{n-1}, 2B_{n-2})$. Hence p is the largest prime of row $n-1$, and $e(n) = 2p$. □

4. Machine verification

Files and audit. The proof is the file `lean/Math0/Frontier/OEIS/A090905.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. The main declaration is `Math0.Frontier.A090905.result`, of type `Math0.Statements.A090905.statement`; all other declarations in the file are private. The file uses no `sorry`, `native_decide`, `axiom` or `set_option`; the finite computations use `decide`, `decide +kernel` and `norm_num`, whose results are checked by the kernel. The file ends with `#math0_settles result statement` (with fully qualified names), a command defined in `lean/Math0/Audit.lean` for statements formalized by math0. It fails unless `statement` is a `Prop`-valued definition without universe parameters, the type of `result` is syntactically `statement` or its negation (up to metadata, without unfolding), and both depend only on the axioms `propext`, `Classical.choice` and `Quot.sound` (which excludes `sorryAx`, `native_decide` and custom axioms). It reports:

```
math0 audit: 'Math0.Statements.A090905.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A090905.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A090905.result' proves 'Math0.Statements.A090905.statement'
```

Toolchain and reproduction. We use Lean 4 [4] `v4.33.1` and Mathlib [3] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [2] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh`: it fetches the Mathlib cache, builds the library with `lake build` (re-running every audit), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and counts the passed statement checks.

References

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